



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

AIOT

13-02-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	B	D	B	B	C	B	D	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	C	D	B	C	C	A	B	C	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	10.00	30.00	2.25 to 2.26	2.00	10.00	10.00	2250.00	60.20	2.00	5.00

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	C	C	D	A	B	C	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	C	D	A	D	C	B	B	C	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	10.00	800.00	7.00	2.00	700.00	8.75	3.00	5.80	108.00	8.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	D	C	D	D	D	A	D	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	A	A	C	B	B	C	A	D	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1.00	255.00	0.00	0.00	4.00	1.00	2019.00	2250.00	7.00	8.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. **Ans (C)**

$$\text{Error} = 1\text{mm} + \frac{25}{100} \times 1\text{mm} = 1.25\text{ mm}$$

$$\begin{aligned} \text{Measured value} &= 18\text{mm} + 34 \times \frac{1}{100} + 1.25 \\ &= 18.34\text{ mm} + 1.25\text{ mm} = 19.59\text{ mm} \end{aligned}$$

2. **Ans (C)**

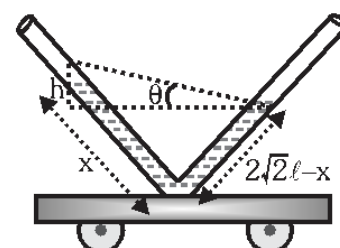
$$\begin{aligned} \text{Translation kinetic energy} &= \frac{3}{2}pV \\ &= \frac{3}{2} \left[\frac{a}{1+1} \right] \times b = \frac{3}{4}ab = 9\text{kJ} \end{aligned}$$

4. **Ans (D)**

The final pressure is $p = \frac{101}{2}\text{ kPa}$; $kx = p\pi^2$ where $x = 0.2$, the compression in the spring and $r = 0.05$ the radius of cylinder.

5. **Ans (B)**

$$\tan \theta = \frac{a}{g} = \frac{h}{\frac{2\sqrt{2}\ell - x}{\sqrt{2}} + \frac{x}{\sqrt{2}}}$$



$$h = \frac{a}{g} \left(\frac{2\sqrt{2}\ell}{\sqrt{2}} \right) = \left(\frac{2a\ell}{g} \right)$$

6. **Ans (B)**

$$\frac{hc}{\lambda} - \phi = \frac{p^2}{2m}$$

$$p = \sqrt{2m \left(\frac{1240}{200} - 6 \right) \times 1.6 \times 10^{-12}}$$

$$= 2 \times 3 \times 4 \times 10^{-26} = 2.4 \times 10^{-25} \text{ kg m/s}$$

$$R_{\max} = \frac{mv}{qB} = \frac{2.4 \times 10^{-25}}{1.6 \times 10^{-19} \times 1} = 1.5 \times 10^{-6} \text{ m}$$

$$X_{\max} = 3 \mu\text{m}$$

7. **Ans (C)**

$$a = v \frac{dv}{ds} = 4 + 3v$$

$$\int_0^v \frac{v dv}{4 + 3v} = \int_0^s ds$$

$$S = 1 - \frac{4}{9} \ell \left(\frac{13}{4} \right)$$

9. **Ans (D)**

$$\frac{L}{R} = 50 \times 10^{-6} \text{ sec}, RC = 20 \times 10^{-3} \text{ sec}$$

$$\Rightarrow \omega = \frac{1}{\sqrt{LC}} = 1000 \Rightarrow f = \frac{1}{2\pi} \frac{1}{\sqrt{LC}} = \frac{1}{2\pi} \text{ kHz}$$

10. **Ans (A)**

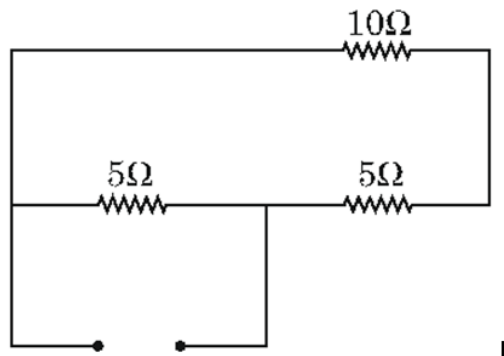
The net magnetic field intensity at the centre 'O' is

$$B_0 = \frac{\mu_0 I}{4\pi R} (-\hat{k}) + \frac{\mu_0 I}{4R} (-\hat{i})$$

$$\therefore B_0 = \frac{\mu_0 I}{4\pi R} \sqrt{1 + \pi^2}$$

11. **Ans (C)**

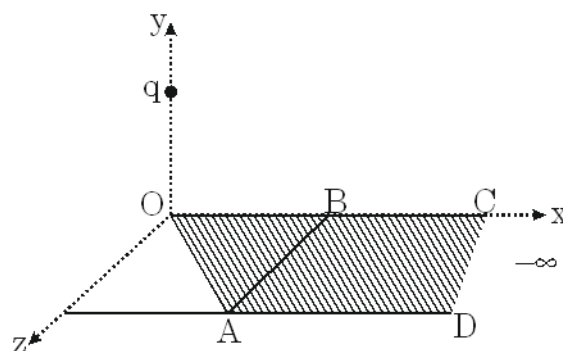
$$x = \frac{15 \times 15}{15 + 5} = \frac{5}{4} \Omega$$



12. **Ans (C)**

$$\phi = \phi_{OAB} + \phi_{ABCD}$$

$$= \frac{q}{48\epsilon_0} + \frac{q}{48\epsilon_0} = \frac{q}{24\epsilon_0}$$



14. **Ans (B)**

$$K = \frac{1}{2} mv^2$$

$$\frac{dK}{dt} = \frac{1}{2} v^2 \frac{dm}{dt} + \frac{1}{2} m \times 2v \frac{dv}{dt}$$

$$= \frac{1}{2} 9t^2 \times 18 + \frac{1}{2} \times 54 \times 2 \times 3t \times 3$$

$$\left. \frac{dK}{dt} \right|_{t=2} = 9 \times 2 \times 18 + 54 \times 18 = 1296 \text{ J/s}$$

17. **Ans (A)**

We know that for a doped semiconductor in thermal equilibrium, we have

$$n_e n_h = n_i^2$$

As per given data,

$$n_i = 1.5 \times 10^{16} \text{ m}^{-3}$$

$$n_h = 4.5 \times 10^{22} \text{ m}^{-3}$$

Thus

$$n_e = \frac{n_i^2}{n_h} = \frac{(1.5 \times 10^{16})^2 \text{ m}^{-6}}{4.5 \times 10^{22} \text{ m}^{-3}} = 5 \times 10^9 \text{ m}^{-3}$$

18. **Ans (B)**

$$\frac{EB}{\mu} = \frac{B^2 C}{\mu} = \text{energy density} \times \text{speed}$$

$$= \text{Energy per unit volume} \times \text{speed}$$

$$= \frac{\text{Energy}}{\text{volume}} \times \text{speed}$$

$$\frac{ML^2 T^{-2}}{L^3} \times LT^{-1} = ML^0 T^{-3}$$

20. **Ans (C)**

$$l = 2 \times \frac{2}{5} mR^2$$

$$l' = 2 \times \frac{2}{5} mR^2 + m \left(\frac{6R}{\sqrt{2}} \right)^2 + \frac{1}{3} m \left(\frac{6R}{\sqrt{2}} \right)^2$$

$$= \left(24 + \frac{4}{5} \right) mR^2$$

PART-1 : PHYSICS

SECTION-II

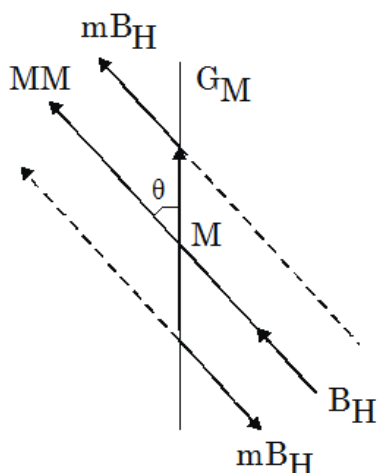
1. **Ans (10.00)**

$$\cos \theta = \frac{r}{R} = \frac{1}{2}$$

$$4\pi r T \cos \theta = \pi r^2 \Delta P$$

2. **Ans (30.00)**

As the compass needle is free to rotate in a horizontal plane and points along the magnetic meridian, so when it is pointing along the geographical meridian, it will experience a torque due to the horizontal component of earth's magnetic field,



$$C = MB_H \sin \theta$$

where θ is the angle between the geographical and magnetic meridians. Substituting the given data,

$$\sin \theta = \frac{(1.2 \times 10^{-3})}{60 \times 40 \times 10^{-6}} = \frac{1}{2}$$

$$\text{i.e., } \theta = \sin^{-1} \left(\frac{1}{2} \right) = 30^\circ$$

3. **Ans (2.25 to 2.26)**

(a) Speed of light in a medium is given by

$$v = \sqrt{\frac{1}{\mu \epsilon}}; \epsilon = K \epsilon_0$$

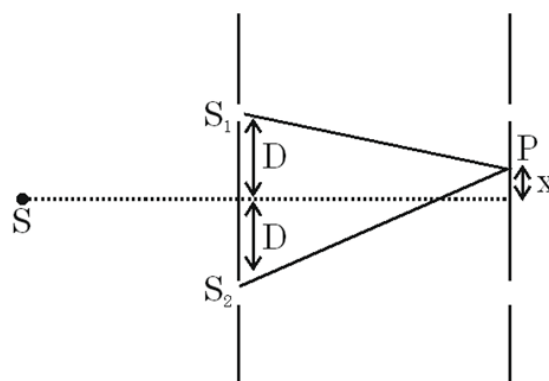
Velocity of light in vacuum,

$$c = \sqrt{\frac{1}{\mu_0 \epsilon_0}}$$

$$v = \frac{1}{\sqrt{\mu_0 K \epsilon_0}} = \frac{c}{\sqrt{K}} = \frac{3.00 \times 10^8}{\sqrt{1.006}}$$

$$= 2.99 \times 10^8 \text{ m/s}$$

4. **Ans (2.00)**



$$\Delta x = \sqrt{(D+x)^2 + D^2} - \sqrt{D^2 + (D-x)^2} = \frac{\lambda}{2}$$

$$\text{and } x = D$$

$$\sqrt{5D^2} - \sqrt{D^2} = \frac{\lambda}{2}$$

$$D = \frac{\lambda}{2(\sqrt{5} - 1)}$$

5. **Ans (10.00)**

Given :

$$\text{Total BW} = 100 \text{ kHz}$$

$$f_{a \max} = 5 \text{ kHz}$$

Any station being modulated by a 5 kHz signal will produce an upper-side frequency 5 kHz above its carrier and a lower-side frequency 5 kHz below its carrier, thereby requiring a bandwidth of 10 kHz.

Thus,

Number of stations accommodated

$$= \frac{\text{Total BW}}{\text{BW per station}} = \frac{100 \times 10^3}{10 \times 10^3}$$

Number of stations accommodated = 10 stations.

6. **Ans (10.00)**

The relative acceleration $a\hat{i} - \frac{3a}{4}\hat{j}$ has a magnitude $\frac{5a}{4} (= a_{\text{rel}})$. the particles must have travelled a distance $\frac{v_0^2}{2a_{\text{rel}}}$ towards each other to reach the minimum separation, d
 $\therefore$ Initial separation = $d + \frac{v_0^2}{2a_{\text{rel}}}$

7. **Ans (2250.00)**

$$f = f_0 \left(\frac{v}{v + v_s \cos \theta} \right)$$

$$f_{\text{min}} = f_0 \left(\frac{v}{v + v_s} \right)$$

$$1800 = 2000 \left(\frac{300}{300 + v_s} \right)$$

$$v_s = 33.33 \text{ m/sec}$$

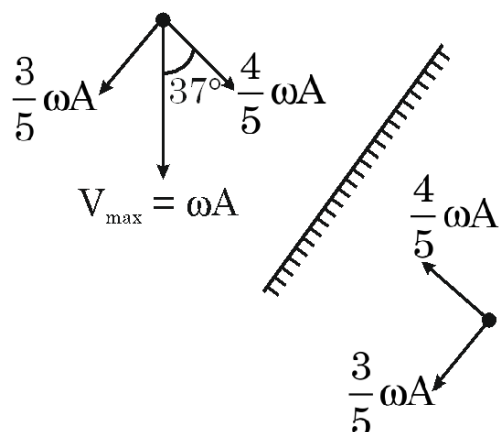
8. **Ans (60.20)**

$$\text{Tension, } T = \frac{3m_1 m_2}{2(m_1 + m_2)} g; v = \sqrt{\frac{T}{\mu}}$$

$$\frac{\delta v}{v} = \frac{1}{2} \frac{\delta T}{T} = \frac{1}{3} \frac{\delta m_2}{m_2} \text{ (where, } m_1 \approx 2m_2 \text{)}$$

10. **Ans (5.00)**

$$V_{\text{ref}} = \frac{8}{5} \omega A = \frac{8}{5} A \sqrt{\frac{K}{M}}$$



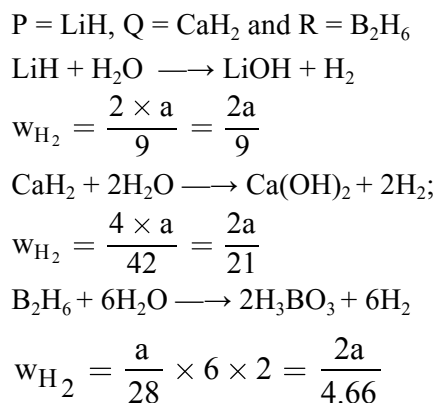
PART-2 : CHEMISTRY

SECTION-I

1. **Ans (B)**

(A) Because of $E^\circ = +1.74 \text{ V}$ for Ce^{+4} to Ce^{+3}
 (B) Due to stability of half filled d-orbitals, Mn^{2+} is more stable than Mn^{3+} and thus its reduction is more spontaneous. Similarly Fe^{3+} is more stable than Fe^{2+} and thus its reduction is less spontaneous.
 (D) Co^{2+} is stable in aqueous solution because it get surrounded and weakly bonded to water molecules. In presence of strong ligands and air it gets oxidised to Co(III) as strong ligands get co-ordinated more strongly with Co(III) . The electronic configuration of Co(II) and Co(III) .

2. **Ans (C)**



3. **Ans (C)**

Rhombic sulphur is more thermodynamically stable than monoclinic sulphur and also oxygen is more stable than ozone.
 $\therefore \Delta H_1 < \Delta H_2 > \Delta H_3 < \Delta H_4$

4. **Ans (C)**

(P) = H_2O , (Q) = CCl_4 , (R) = SiCl_4 , (S) = H_2 ,
 (T) = NH_3 , (U) = AsH_3 , (V) = HF ,
 (X) = CH_4 & (Y) = Cl_2

5. **Ans (D)**

A = CHF_3 , B = NaOI , C = CHI_3 ,
D = HCOONa , P = CH_3OH , Q = NaI

7. **Ans (B)**

PAN- peroxy acetyl nitrate is most harmful.

8. **Ans (C)**

(A) Initially, mms of $\text{OH}^- = 100$,
mms of $\text{H}^+ = 10$, $[\text{OH}^-] = 90/200$
 $\Rightarrow \text{pOH} = 0.346$
Finally, mms of $\text{OH}^- = 100$,
ms of $\text{H}^+ = 30$, $[\text{OH}^-] = 70/200$
 $\Rightarrow \text{pOH} = 0.456$
So, change in pH = 0.111 (small change)

(B) After initial reaction,
mms of $\text{HS}^- = 20$, mms of $\text{H}_2\text{S} = 20$
(its a buffer of $\text{HS}^-/\text{H}_2\text{S}$)
So, change in pH will be very small.

(C) Initially, mms of $\text{OH}^- = 10$,
mms of $\text{H}^+ = 0$, $[\text{OH}^-] = 10/200$
 $\Rightarrow \text{pH} = 12.7$
Finally, mms of $\text{OH}^- = 10$,
mms of $\text{H}^+ = 20$, $[\text{H}^+] = 10/200$
 $\Rightarrow \text{pH} = 1.3$
So, change in pH = 11.4 (very large)

(D) Initially, mms of $\text{AcOH} = 30$,
mms of $\text{AcO}^- = 20$,
mms of $\text{H}_2\text{S} = 20$ (its a buffer)
So, change in pH will be very small.

9. **Ans (C)**

Radial node occurs where probability of finding e^- is zero.

$$\therefore \psi^2 = 0 \text{ or } \psi = 0$$

$$\therefore 6 - 6\sigma + \sigma^2 = 0; \sigma = 3 \pm \sqrt{3}$$

$$\text{For max. distance } r = \frac{3}{2} \frac{(3 + \sqrt{3})a_0}{Z}$$

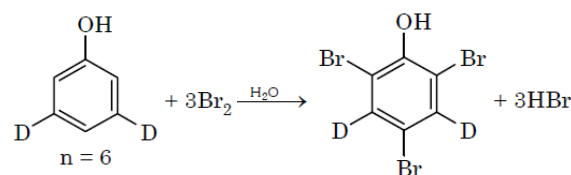
11. **Ans (B)**

$$\ln \frac{P_1}{P_2} = \frac{\Delta \ominus H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\text{or, } \ln \frac{0.4}{59.2} = \frac{-16.628 \times 10^3}{8.314} \left(\frac{1}{200} - \frac{1}{T} \right)$$

$$\Rightarrow T = 400 \text{ K}$$

12. **Ans (C)**



Moles of (Y) = 1

Moles of $\text{NaOH} = 0.5$

$\text{pH} = \text{pKa} = 10$

13. **Ans (D)**



$$\begin{array}{ccc} 29.6 & 15.4 \text{ gm} & 4.48 \\ & & \hline & & 22.4 \end{array} \text{ moles} = 0.2 \text{ moles}$$

$$\Rightarrow n_{\text{MX}_3} = 0.2 = n_{\text{MX}}$$

$$\Rightarrow \frac{29.6}{m + 3x} = 0.2$$

$$m + 3x = 148 \quad \dots(1)$$

$$\text{Also } \frac{15.4}{m + x} = 0.2$$

$$m + x = 77 \quad \dots(2)$$

Solving (1) and (2)

$$x = 35.5 \quad 8m = 41.5$$

$\therefore x$ is chlorine

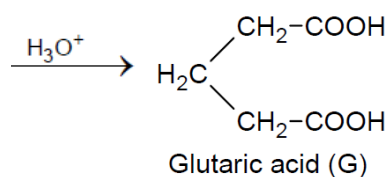
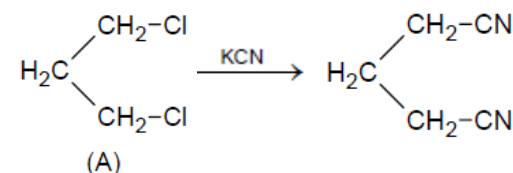
14. **Ans (A)**

$\text{CH}_3\text{-CH}_2\text{-SH} \quad \text{pKa} = 10.5$

$\text{Ph-SH} \quad \text{pKa} = 7.8$

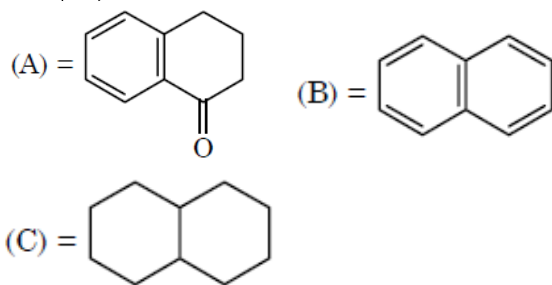
$\text{Et-OH} \quad \text{pKa} = 15.9$

16. **Ans (C)**

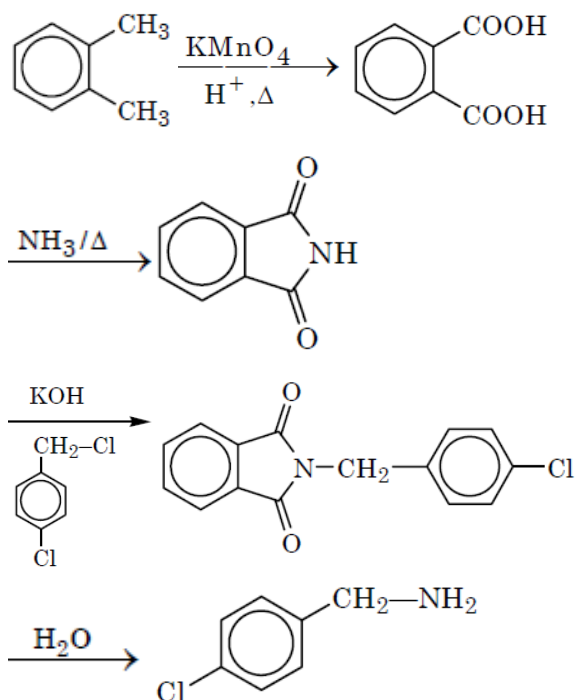


Hence (C) is correct.

17. Ans (B)



19. Ans (C)



20. Ans (C)

Gas (Z) = B₂H₆

$$V = \frac{ZnRT}{P} = 2.24 \text{ mL}$$

PART-2 : CHEMISTRY

SECTION-II

1. Ans (10.00)

P = 6, Q = 4, R = 0.

2. Ans (800.00)

$$W_{\text{total}} = W_{AB} + W_{BC} + W_{CA}$$

$W_{AB} = 0$ (isochoric)

$$W_{BC} = \frac{nR\Delta T}{\gamma - 1} \quad (\text{Polytropic } PV^{1/2} = \text{constant})$$

$$= \frac{1 \times 2 \times 600}{-1/2} = -2400 \text{ Cal}$$

$W_{CA} = -nR\Delta T$ (isobaric)

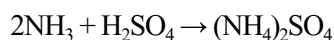
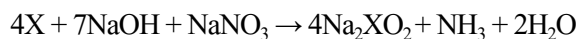
$$= -1 \times 2 \times -800 = 1600 \text{ Cal}$$

$$W_{\text{total}} = 0 - 2400 + 1600 = -800 \text{ Cal}$$

3. Ans (7.00)

iii, iv, v, viii, ix, x, xi

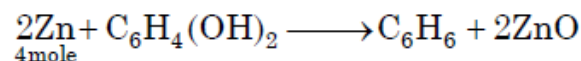
4. Ans (2.00)



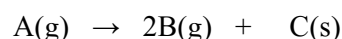
$$4 \times 10^{-2} = \frac{2.616}{A} \quad (A = \text{Atomic mass of 'X'})$$

$$\therefore A = 65.4$$

$\therefore X$ is Zn



5. Ans (700.00)



$$t = 0 \quad 400 \quad 0 \quad 0$$

$$t = t \quad (400 - p') \quad 2p' \quad 0$$

Since the reaction is of first order

$$k = \frac{2.303}{t} \log \frac{400}{(400 - p')}$$

$$\frac{0.6932}{24} = \frac{2.303}{48} \log \frac{400}{(400 - p')}$$

On solving $p' = 300$

After 48 minimum the pressure of the reaction vessel is $(400 + p') = 700 \text{ mm Hg}$.

6. Ans (8.75)

$$P_0 = x_A P_A^0 + x_B P_B^0$$

$$= \frac{2}{8} \times 300 + \frac{6}{8} \times 500 = 450 \text{ mm}$$

$$\text{RLVP} = \frac{P_0 - P_5}{P_0} = \frac{450 - 420}{450} = \frac{\frac{32}{70}i}{\frac{32}{70}i + 8}$$

$$\Rightarrow i = 1.25 = 1 + (n + 1)\alpha \Rightarrow \alpha = 0.2$$

$$n_{\text{Cl}^-} \text{ produced} = \frac{32}{70} \times 0.25 = \frac{4}{35} = x$$

$$\frac{1}{x} = \frac{35}{4} = 8.75$$

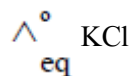
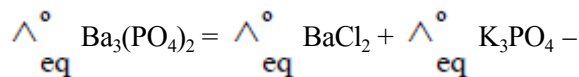
8. Ans (5.80)

$$x = 5$$

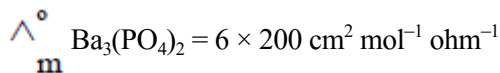
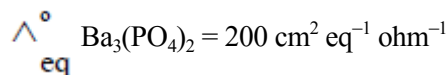
$$y = 2$$

$$z = 5$$

9. Ans (108.00)



$$= 160 + 140 - 100$$



$$\overset{\circ}{\text{m}} = K \times \frac{1000}{\text{solubility}}$$

$$1200 = 12 \times 10^{-6} \times \frac{1000}{\text{solubility}}$$

$$s = 10^{-5}$$

$$K_{sp} = (3s)^3 (2s)^2$$

$$K_{sp} = 3^3 2^2 s^5$$

$$K_{sp} = 108 \times 10^{-25}$$

10. Ans (8.00)

P $\Rightarrow$ Pb, Hg, Cu

Q $\Rightarrow$ Al, Na, Mg

R $\Rightarrow$ Zn, Hg

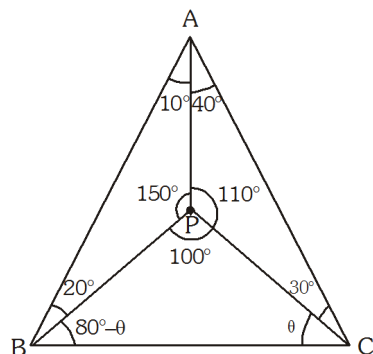
So, P = 3, Q = 3 & R = 2

So answer is 3 + 3 + 2 = 8

PART-3 : MATHEMATICS

SECTION-I

1. Ans (D)



$$\frac{PA}{PB} \cdot \frac{PB}{PC} \cdot \frac{PC}{PA} = 1$$

$$\frac{\sin 20^\circ}{\sin 10^\circ} \times \frac{\sin \theta}{\sin (80^\circ - \theta)} \times \frac{\sin 40^\circ}{\sin 30^\circ} = 1$$

$$\theta = 20^\circ$$

2. Ans (B)

$$4 \leq \text{LHS} \leq 16$$

$$2 \leq \text{RHS} \leq 4$$

Hence equality can occur at 4, which is possible if $x = \pi, 3\pi, 5\pi \Rightarrow 3$ solutions

3. Ans (D)

For domain of $f(x)$, we must have

$$-1 \leq [x] \leq 1 \Rightarrow -1 \leq x < 2,$$

so set $A = [-1, 2)$

$$f(x) = \frac{2}{\pi} \left(\sin^{-1} [x] + \frac{\pi}{2} \right)$$

$$\left(\text{As } \tan^{-1} [x] + \cot^{-1} [x] = \frac{\pi}{2} \forall x \in A \right)$$

set $B = \{0, 1, 2\} = \text{Range of } f(x)$

$$\text{Now } A \cup B = [-1, 2) \cup \{0, 1, 2\} = [-1, 2]$$

Hence number of integers in $(A \cup B) = 4 = m$

Given expression

$$\begin{aligned} &= \left\{ \frac{\pi}{2} - \cos^{-1}(\cos 2) \right\} - \left\{ \frac{\pi}{2} - \sin^{-1}(\sin 2) \right\} + \left\{ \frac{\pi}{2} - \cot^{-1}(\cot 4) \right\} \\ &- \left\{ \frac{\pi}{2} - \tan^{-1}(\tan 4) \right\} + \left\{ \frac{\pi}{2} - \operatorname{cosec}^{-1}(\operatorname{cosec} 6) \right\} - \left\{ \frac{\pi}{2} - \sec^{-1}(\sec 6) \right\} \\ &= \sin^{-1}(\sin 2) - \cos^{-1}(\cos 2) + \tan^{-1}(\tan 4) - \cot^{-1}(\cot 4) \\ &+ \sec^{-1}(\sec 6) - \operatorname{cosec}^{-1}(\operatorname{cosec} 6) \\ &= (\pi - 2) - 2 + (4 - \pi) - (4 - \pi) + (2\pi - 6) - (6 - 2\pi) \\ &= \pi - 4 + 4\pi - 12 = 5\pi - 16 = 5\pi - 4m \end{aligned}$$

4. Ans (C)

$$6 \leq 2 \left| \frac{3x + 4y - 7}{5} \right| + 3 \left| \frac{4x - 3y - 1}{5} \right| \leq 12$$

$$\Rightarrow 6 \leq 2|x| + 3|y| \leq 12$$

$2|x| + 3|y| \leq 12$ is the area inside a rhombus

$$\text{having area} = \frac{2(12)^2}{2 \times 3}$$

Similarly $2|x| + 3|y| \geq 6$ is area outside a

$$\text{rhombus with area} = \frac{2(6)^2}{2 \times 3},$$

so reqd ans is 36

5. Ans (D)

Let $(\alpha, 0)$ be the midpoints of the chord. The other end of the chord is $(2\alpha - p, -q)$ which lies on the circle :

$$\Rightarrow (2\alpha - p)^2 + q^2 = p(2\alpha - p) - q^2$$

$$\Rightarrow 2\alpha^2 - 3p\alpha + p^2 + q^2 = 0$$

For two values of α , we have

$$9p^2 > 8(p^2 + q^2) \text{ or } p^2 > 8q^2$$

6. **Ans (D)**

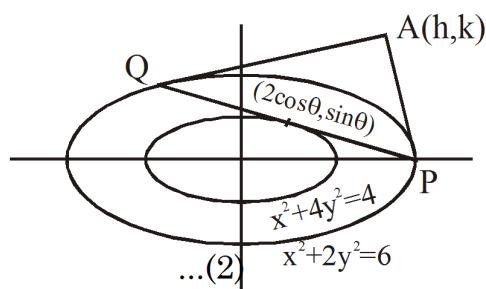
Parabola is $(x-2)^2 = 8(y-1)$

Now equation of normal having slope $-\frac{1}{2}$ is

$$(y-1) = -\frac{1}{2}(x-2) + 4 + \left(\frac{2}{(1/4)}\right)$$

$$\Rightarrow x + 2y = 28$$

7. **Ans (D)**



Equation of PQ

$\Rightarrow$ Chord of contact

$$\Rightarrow hx + 2yk = 6 \quad \dots(1)$$

Equation of PQ

$\Rightarrow$ Equation tangent

$$\Rightarrow \frac{x \cos \theta}{2} + \frac{y \sin \theta}{1} = 1$$

Compare (1) and (2)

$$\frac{h \cdot 2}{\cos \theta} = \frac{2k}{\sin \theta} = 6$$

$$\cos \theta = \frac{h}{3}, \sin \theta = \frac{k}{3}$$

$\Rightarrow x^2 + y^2 = 9$ which is director circle of

$$x^2 + 2y^2 = 6 \Rightarrow AP \perp AQ$$

8. **Ans (A)**

Let the perpendicular lines be co-ordinate axes

$$\therefore f^2 - c = k_1 \text{ \& } g^2 - c = k_2$$

$$\Rightarrow f^2 - g^2 = k_1 - k_2 \therefore x^2 - y^2 = k$$

$$\therefore e = \sqrt{2}$$

9. **Ans (D)**

$$\alpha^5 = 1 \Rightarrow \frac{1}{\alpha} = \alpha^4$$

$$\Rightarrow \left| 1 + \alpha + \alpha^2 + \alpha^3 - \frac{1}{\alpha} \right| = \left| -\alpha^4 - \alpha^4 \right| = 2$$

Hence answer is (D).

10. **Ans (C)**

As the difference of roots of

$f(x) = 0$ and $g(x) = 0$ are equal

$$\text{then } \frac{\sqrt{D_1}}{|a_1|} = \frac{\sqrt{D_2}}{|a_2|}$$

$$\Rightarrow p^2 - 4q = r^2 - 4s$$

Now minimum value of

$$f(x) = -\frac{D_1}{4a_1} = \frac{-(p^2 - 4q)}{4} = -\frac{1}{4}$$

$$\Rightarrow \frac{4s - r^2}{4} = -\frac{1}{4} \Rightarrow r^2 - 4s = 1 \quad \dots(1)$$

Least value of $g(x)$ occurs at

$$x = -\frac{b_2}{2a_2} = -\frac{r}{2} = \frac{7}{2}$$

$$\Rightarrow s = 12 \Rightarrow g(x) = x^2 - 7x + 12 = 0$$

$$\Rightarrow x = 3, 4$$

11. **Ans (C)**

$$-1 \leq \log_2 \left(\frac{x^2}{2} \right) \leq 1$$

$$\Rightarrow \frac{1}{2} \leq \frac{x^2}{2} \leq 2$$

$$\Rightarrow 1 \leq x^2 \leq 4$$

$$\Rightarrow x \in [-2, -1] \cup [1, 2]$$

12. **Ans (A)**

For non-trivial solution

$$\begin{vmatrix} \sin \theta & -2 \cos \theta & -a \\ -1 & 1 & 1 \\ 1 & 2 & 1 \end{vmatrix} = 0 \Rightarrow \sin \theta + 4 \cos \theta = 3a$$

$$-\frac{\sqrt{17}}{3} \leq a \leq \frac{\sqrt{17}}{3}$$

so 3 integral values.

13. **Ans (A)**

$$S = \frac{3}{5} + \frac{10}{5^2} + \frac{21}{5^3} + \frac{36}{5^4} + \frac{55}{5^5} + \dots \infty$$

$$\frac{S}{5} = \frac{3}{5^2} + \frac{10}{5^3} + \frac{21}{5^4} + \frac{36}{5^5}$$

Subtracting

$$\frac{4}{5}S = \frac{3}{5} + \frac{7}{5^2} + \frac{11}{5^3} + \frac{15}{5^4} + \frac{19}{5^5}$$

$$\frac{4}{5^2}S = \frac{3}{5^2} + \frac{7}{5^3} + \frac{11}{5^4}$$

subtracting

$$\frac{16}{25}S = \frac{3}{5} + \frac{4}{5^2} + \frac{4}{5^3} + \frac{4}{5^4} + \dots$$

$$= \frac{3}{5} + \frac{\frac{4}{5^2}}{1 - \frac{1}{5}} = \frac{3}{5} + \frac{1}{5}$$

$$\frac{16}{25}S = \frac{4}{5}$$

$$S = \frac{5}{4}$$

$$\Rightarrow 4S = 5$$

14. **Ans (C)**

$$\text{Let } xyz = 72 = 2^3 \times 3^2$$

then number of solutions are

$${}^{3+3-1}C_{3-1} \times {}^{2+3-1}C_{3-1}$$

$$= {}^5C_2 \times {}^4C_2 = 10 \times 6 = 60$$

15. **Ans (B)**

$$|PQ| = k^3 \Rightarrow |P| = 2k \neq 0$$

$\Rightarrow P$ must be invertible

$$\Rightarrow Q = kp^{-1} = k \frac{(\text{adj } P)}{|P|} = \frac{(\text{adj } P)}{2}$$

$$\text{now } q_{23} = -\frac{k}{8} = \frac{-(3\alpha + 4)}{2} \dots (i)$$

$$|p| = 12\alpha + 20 = 2k \dots (ii)$$

$$\text{Solving (1) \& (2) } \alpha = -1, k = 4$$

$$\Rightarrow \alpha^2 + k^2 = 17$$

16. **Ans (B)**

$$X : 51, 52, 53 \dots 100$$

$$Y : 151, 152, 153 \dots 200$$

$$V_x = V_y \text{ hence } \lambda = 1$$

17. **Ans (C)**

$$P(A) = \frac{5}{9};$$

$$P(A \cap B) = \frac{1}{6} = \frac{5}{9} \times \frac{3}{10} = P(A)P(B)$$

18. **Ans (A)**

$$x^2 + 2x + 3 + \sin \pi x = (x+1)^2 + 2 + \sin \pi x > 1$$

$$\therefore f(x) = 1 \forall x \in \mathbb{R}$$

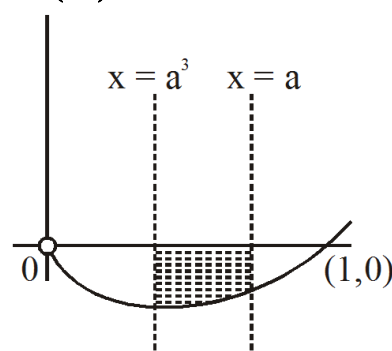
19. **Ans (D)**

$f'(x)$ is -ve

$\Rightarrow f(x)$ is decreasing function

Hence maximum value is $-\frac{1}{8}$ at $x = 0$

20. **Ans (C)**



From figure,

$$\Delta(a) = - \int_{a^3}^a x \ln x \, dx = \int_a^{a^3} x \ln x \, dx$$

$$\Delta'(a) = a^3 \ln(a^3) \cdot 3a^2 - a \ln a \cdot 1$$

(using Newton Leibnitz Rule).

$$\text{So } \Delta'(a) = a \ln a (9a^4 - 1) = 0$$

$$\text{gives } a = \frac{1}{\sqrt{3}} \quad \Delta'(a) \text{ changes sign}$$

from positive to negative while passing through

$$\frac{1}{\sqrt{3}} \text{ so it is a maxima.}$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (1.00)**

$p \Rightarrow q$ is logically equivalent to $\sim q \Rightarrow \sim p$
 $\therefore (p \Rightarrow q) \Leftrightarrow (\sim q \Rightarrow \sim p)$ is a tautology
 but not a contradiction
 only (S-2) is false

2. **Ans (255.00)**

$n(A \times B) = 8$
 number of relations $= 2^8 - 1 = 255$

3. **Ans (0.00)**

$$\begin{aligned} & \int \frac{2x^7 + 3x^2}{x^{10} - 2x^5 + 1} dx \\ &= \int \frac{x^6 \left(2x + \frac{3}{x^4}\right)}{x^6 \left(x^4 - \frac{2}{x} + \frac{1}{x^6}\right)} dx \\ &= \int \frac{2x + \frac{3}{x^4}}{\left(x^2 - \frac{1}{x^3}\right)^2} dx \end{aligned}$$

$$\begin{aligned} \text{Let } x^2 - \frac{1}{x^3} &= t \Rightarrow \left(2x + \frac{3}{x^4}\right) dx = dt \\ &= \int \frac{dt}{t^2} = -\frac{1}{t} + c = -\frac{x^3}{x^5 - 1} + c \\ &= \frac{-x^3(x^5 - 1)}{(x^5 - 1)^2} + c = \frac{x^3 - x^8}{(x^5 - 1)^2} + c \end{aligned}$$

4. **Ans (0.00)**

$$\begin{aligned} & \lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{{}^nC_r}{n^r} \cdot \left(\int_0^1 x^{r+3} dx \right) \\ &= \int_0^1 \left(\lim_{n \rightarrow \infty} \sum_{r=0}^n {}^nC_r \cdot \left(\frac{x}{n}\right)^r \cdot x^3 \right) dx \\ &= \int_0^1 \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n \cdot x^3 dx \\ &= \int_0^1 e^x \cdot x^3 dx = 6 - 2e \end{aligned}$$

5. **Ans (4.00)**

$$\begin{aligned} L &= \lim_{x \rightarrow 0} \frac{x^4 + 3(a^2 - \sqrt{a^4 + x^4})}{x^8} \left(\frac{0}{0}\right) \\ &= \lim_{x \rightarrow 0} \frac{(x^4 + 3a^2) - 3\sqrt{a^4 + x^4}}{x^8} \left(\frac{0}{0}\right) \\ &= \lim_{x \rightarrow 0} \frac{(x^4 + 3a^2)^2 - 9(a^4 + x^4)}{x^8 ((x^4 + 3a^2) + 3\sqrt{a^4 + x^4})} \end{aligned}$$

(on rationalising)

$$= \frac{1}{6a^2} \lim_{x \rightarrow 0} \left(\frac{x^8 + 6a^2x^4 - 9x^4}{x^8} \right)$$

$$L = \frac{1}{6a^2} \lim_{x \rightarrow 0} \left(1 + \frac{(6a^2 - 9)}{x^4} \right)$$

$$\text{Clearly } 6a^2 - 9 = 0 \Rightarrow a^2 = \frac{9}{6} = \frac{3}{2}$$

$$\therefore a = \sqrt{\frac{3}{2}} \quad (\text{As } a > 0)$$

$$\text{Also } L = \frac{1}{6a^2} = \frac{1}{9}$$

6. **Ans (1.00)**

$$\therefore \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$$

$$\begin{aligned} \frac{d^2x}{dy^2} &= \frac{d}{dy} \left(\frac{1}{\frac{dy}{dx}} \right) = \frac{d}{dx} \left(\frac{1}{\frac{dy}{dx}} \right) \cdot \frac{dx}{dy} \\ &= \frac{-1}{\left(\frac{dy}{dx}\right)^2} \cdot \frac{d^2y}{dx^2} \cdot \frac{dx}{dy} = \frac{-1}{\left(\frac{dy}{dx}\right)^3} \cdot \frac{d^2y}{dx^2} \end{aligned}$$

$$\text{From the given is } \frac{d^2y}{dx^2} \left(1 - \frac{1}{\left(\frac{dy}{dx}\right)^3} \right) = 0$$

$$\Rightarrow \frac{d^2y}{dx^2} = 0 \text{ or } \frac{dy}{dx} = 1$$

$$\Rightarrow y = ax + b : a \neq 0$$

7. **Ans (2019.00)**

We have $y = 2019x^2$

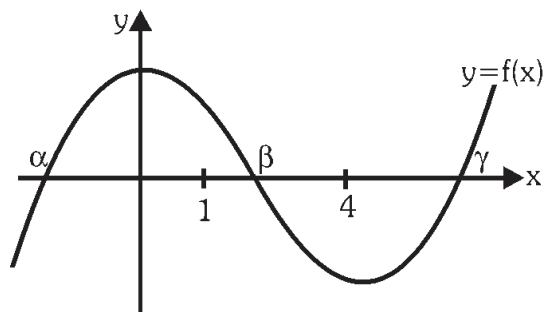
$$(AB)^2 = (x_B - x_A)^2 + (2019x_B^2 - 2019x_A^2)^2 \\ = 2019^2 + 1$$

or

$$(x_B - x_A)^2 + 2019^2(x_B^2 - x_A^2)^2 = 2019^2 + 1$$

differentiating w.r.t. x_A and denoting

$$\frac{dx_B}{dx_A} = D$$



$$2(x_B - x_A)(D - 1) + 2 \cdot 2019^2$$

$$(x_B^2 - x_A^2)(2x_B D - 2x_A) = 0$$

put $x_A = 0$; $x_B = 1$

$$2(1 - 0)(D - 1) + 2 \cdot 2019^2$$

$$(1 - 0)(2D - 0) = 0$$

$$2D - 2 + 4 \cdot 2019^2 D = 0$$

$$\Rightarrow D = \frac{1}{2 \cdot 2019^2 + 1}$$

8. **Ans (2250.00)**

∴ Speed of the truck is x km/h

$$\therefore \text{Time taken to cover 500 km is } t = \left(\frac{500}{x} \right) \text{ h}$$

Total cost for a trip of 500 km (in Rs) is = cost of diesel + Payment to the driver

$$C \text{ (say)} = \frac{1}{300} \left(\frac{900}{x} + x \right) \times 500 \times \frac{400}{100} + \frac{500}{x} \times 1.5$$

$$\Rightarrow C = \frac{1350}{x} + \frac{2}{3}x$$

$$\therefore \frac{dC}{dx} = -\frac{1350}{x^2} + \frac{2}{3}; \text{ for max or min of } C$$

$$\frac{dC}{dx} = 0$$

$$\Rightarrow x = 45 \text{ km/h}$$

$$\text{Now, } \frac{d^2C}{dx^2} = \frac{2700}{x^3}$$

$$\therefore \left| \frac{d^2C}{dx^2} \right|_{x=45} = \frac{2700}{(45)^3} > 0$$

Total cost C is minimum where $x = 45$ km/h

$$\therefore \lambda = 45$$

$$\text{Then, } 50\lambda = 50 \times 45 = 2250$$

9. **Ans (7.00)**

$$\frac{y \cdot y'' - (y')^2}{(y')^2} = \frac{(2x-1)(y')^2}{(y')^2}$$

$$- \int d \left(\frac{y}{y'} \right) = \int (2x-1) dx$$

$$y = e^{\frac{4x}{2x-1}}$$

10. **Ans (8.00)**

$$\text{Volume of the parallelopiped} = \left| [\vec{a} \vec{b} \vec{c}] \right|$$

⇒ (area of the base parallelogram)

$$h = \left| [\vec{a} \vec{b} \vec{c}] \right|$$

$$\Rightarrow |\vec{a} \times \vec{b}| h = \left| [\vec{a} \vec{b} \vec{c}] \right|$$

$$\Rightarrow \sqrt{38} h = 4$$

$$\Rightarrow h = \frac{4}{\sqrt{38}}$$